

1-C:

A - This is not a negative externality because the reduced profit is a direct result of the market competition. This may represent a reduction in private benefits as the impact is internal to the firm, not external.

B. This is not a negative externality because the reduced funding is a direct result of a policy decision and does not represent the negative impacts borne by the third party.

C. This is a negative externality because the increased noise levels are a harmful effect of airport expansion. The noise pollution is not reflected in the market price of airline tickets, which means that it is not internalized by the competitive free market.

D. This is not a negative externality because the increased cost represents a private cost to the firm, not an external impact on third parties.

2-C:

A – Point “A” represents a normal profit situation because Average costs are equal to Average revenue.

B – Point “B” represents a productively efficient situation because marginal cost is equal to Average cost which represents that Average total cost has been minimized.

C – Point “C” represents an allocatively efficient point because marginal cost pricing is done, or in other words, marginal cost is equal to price.

D – Point “D” represents a sales revenue maximizing point because $MR=0$.

3-C:

A - While educators may be scarce, this doesn't necessarily mean training is not socially optimal.

B. Training is not typically considered a public good, as it's not non-rivalrous and non-excludable; instead, it exhibits the characteristics of excludability and rivalry.

C – Since the social benefit is greater than the individual's benefit, which represents the private benefit gained by the individual, it can be said that there is existence of “external benefits”. In a mixed economy, individuals and firms may not invest in optimal levels of training because they only consider their private benefits, thereby the external benefits that could have been gained are translated into deadweight loss of economic welfare due to under-allocation of resources.

D. While training may be a significant cost for firms, this doesn't explain why the amount of training is not socially optimal.

4-A:

Hint: "Price fixing"

- ➔ **Collusion: Fixing prices** on some routes to maximize revenue indicates an agreement among competitors to set prices.
- ➔ **Limit pricing:** Lowering prices on other routes to prevent smaller competitors from entering the market represents limit pricing, which is aimed to deter potential entrants from entering.

B. **Price leadership** refers to a single firm who is the brand leader with the highest market share setting prices, which is followed by others, which isn't the case here.

C. **Price discrimination** is charging different prices for the same product to different customers; the reasons of which, do not depend upon cost but instead depend on price elasticity of demand. This is also not the case here.

D. **X-inefficiency** refers to inefficiencies in production arising from a lack of competitive pressures, not related to the anti-competitive practices mentioned.

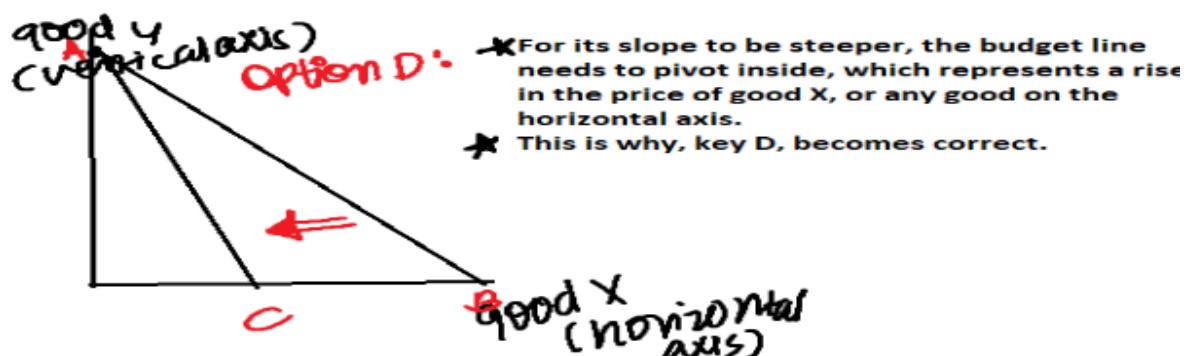
5-D:

A - If the consumer's income fell, the budget line would have shifted leftwards parallelly, but the slope would have remained unchanged, since **parallel lines have the same gradient**.

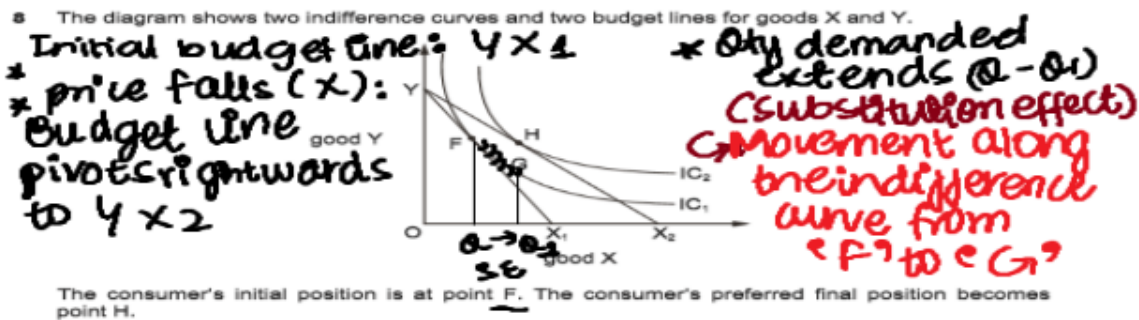
B - If the consumer's income rose, the budget line would have shifted rightwards parallelly, but the slope would have remained unchanged, since **parallel lines have the same gradient**.

C - If the price decreased for the good on the horizontal axis, the budget line would have tilted outwards, and as a result, the gradient or the slope would have become flatter.

D -



6- C:



- A- The income effect of a price fall for X would be from "G to H"
- B- The price effect represents the total effect, which comprises of both: income effect and the substitution effect, and is shown from "F" to "H"
- C- It can be seen in the diagram above that the substitution effect for a fall in the price of X is shown through the movement along the indifference curve from "F to G"
- D- This is incorrect as the quantity would have contracted if the price had risen for X.

7- A:

output (chairs)	total cost (\$)
0	50
1	60
2	64
3	77
4	94
5	114

→ Fixed cost is also incurred when the output is zero; therefore, "Fixed cost = 50"

Variable cost = $TC - FC$	Average variable cost =
* $VC = 94 - 50$ $VC = 44$	* $\frac{VC}{Qty} \Rightarrow \frac{44}{4} \Rightarrow 11$

8-A:

A: A firm achieves sales maximization when it charges the minimum price possible while avoiding losses. This means that it occurs when Average revenue (price) is equal to Average costs, or in other words, when normal profits are made. Therefore, option A, becomes correct.

B : When marginal cost is equal to Average cost, this represents that Average total cost has been minimized, which also means that productive efficiency is achieved.

C: When marginal cost is equal to Average revenue, this represents allocative efficiency.

D: When marginal revenue is equal to zero, this represents that sales revenue has been maximized.

9-B:

A: This is a typical feature of monopolistic competition. Since firms produce differentiated products, they compete on the basis of other attributes like quality, design, and brand image. Advertising is a key tool for firms to differentiate their products and attract customers.

B. Price leadership is a characteristic of oligopoly markets, where a few large firms dominate the market and one firm who is the brand leader sets prices that others follow. In monopolistic competition, there are many firms, and no single firm has significant market power to influence prices for the entire market .This is why, this becomes correct.

C. This is another key feature of monopolistic competition. Firms produce differentiated products to distinguish themselves from competitors and attract customers. Product differentiation can be achieved through various means, such as quality, design, branding, and packaging.

D. In monopolistic competition, there are weak barriers to entry, and firms can enter or exit the market freely. In the short-run when firms earn supernormal profits due to product differentiation, firms enter because of weak barriers to entry and exit. As a result of this, the demand curve faced by an individual firm shifts to the left, because of which, prices fall. This leads to a situation where profits are normalized in the long run.

10-D:

A - This option is incorrect because the principal-agent problem is not caused by managers being prevented from becoming shareholders.

B - This option is incorrect because a cartel behavior is a form of collusion between competing firms, whereas the principal-agent problem occurs within a single firm between owners and managers.

- C - This option is incorrect because market dominance does not necessarily lead to a principal-agent problem. The issue arises from the separation of ownership and control, not from a firm's market position.
- **D** - This option is correct because the principal-agent problem occurs when the goals and objectives of owners (principals) differ from those of managers (agents), leading to potential conflicts of interest. This is because, the managers prefer sales revenue maximization since their salaries and bonuses are linked to revenues and not profits, as compared to shareholders who prefer profit maximization.

11-C:

C - The kinked demand curve is a graphical representation of an oligopolistic market, where the demand curve for an individual firm is assumed to be highly elastic (sensitive to price changes) above the current market price and highly inelastic (less sensitive to price changes) below the current market price. This kinked shape reflects the assumption that firms in an oligopoly are interdependent, meaning that they take into account the potential reactions of their competitors when making pricing decisions. **This is why, prices remain unchanged at the kink.**

- A. Price discrimination: This refers to a pricing strategy where firms charge different prices to different customers for the same product, the reasons of which do not depend upon costs but instead depend upon price elasticity of demand.
- B. Price leadership by the dominant firm: This occurs when a dominant firm sets the market price, and other firms follow. While this can happen in oligopoly, it's not the specific feature assumed in the kinked demand curve.
- D. Collusion between firms: Collusion refers to explicit agreements between firms to fix prices or output. While collusion can occur in oligopoly, the kinked demand curve assumes interdependence between firms (in a non-collusive oligopoly), not collusion.

12- B:

A - For 2014, Average value= $\text{€}980 \text{ billion} \div 17,000 = \text{approximately } \text{€}0.057 \text{ billion}$

And similarly, for 2017:

$\text{€}980 \text{ billion} \div 18,000 = \text{approximately } \text{€}0.054 \text{ billion}$

This represents that the average value of a merger was higher in 2014 than in 2017.

B - The correct answer, option B, can be verified by calculating the difference in the number of mergers between Europe and North America for 2014 and 2016:

2014: 17,000 (Europe) - 14,000 (North America) = 3,000

2016: 18,100 (Europe) - 15,100 (North America) = 3,000

- C - This is incorrect, as the values are in different currencies (Euros and US Dollars). We can't compare them directly.
- D: This is incorrect, as the number of mergers in North America was higher in 2017.

13-D:

A. Game theory is the study of strategic decision making in situations where outcomes depend on the actions of multiple parties. However, in this scenario, the charity's action of leaving a note is not a strategic decision made in response to other charities' actions, but rather a unilateral gesture of gratitude.

B. Monopoly behavior refers to the actions of a single firm that has complete control over a market. However, in this scenario, there are multiple charities competing for donations, so it's not a monopoly situation.

C. Non-price competition refers to competition among firms based on factors other than price, such as quality, service, or branding. The charity is not trying to attract donors away from other charities, but simply expressing thanks for the donation

D. Nudge theory refers to the use of subtle cues and incentives to influence people's behavior in predictable ways, without limiting their freedom of choice. The charity's note is a subtle cue that aims to influence the householder's behavior (encouraging them to donate again). The note is a gentle, non-intrusive message that shapes behavior, making nudge theory the best fit.

14- B:

A – An increase in barriers to entry can lead to market failure by limiting competition and creating monopolies.

B - Building regulations are laws and standards that govern the construction and maintenance of buildings. These regulations can prevent market failure by ensuring safety standards, protecting the environment and preventing negative externalities (e.g., pollution). By setting minimum standards, building regulations can prevent firms from engaging in harmful practices that lead to market failure.

C- While firm growth can be beneficial, it can also lead to market failure if it results in monopolies or exploitation of market power.

D- Persuasive advertising can lead to market failure by creating artificial demand after misleading consumers.

15- D:

Note: Equity is different from equality. Equality aims to treat everyone the same, while equity aims to provide everyone with what they need to thrive, thus making it a fair and just approach.

This means that equity believes in doing what is just, fair and right from the society's point of view. It recognizes that each individual has a different set of needs and so an equal distribution of income may not be fair.

A, B, C : When all individuals have equal job opportunities, all workers are paid the same wages, and when incomes are equally distributed, this represents the concept of equality.

D: Since equity is based on the idea of fairness, an equitable distribution of income will occur when the society believes that the distribution of income is fair. This explains why key D will be correct.

16-C:

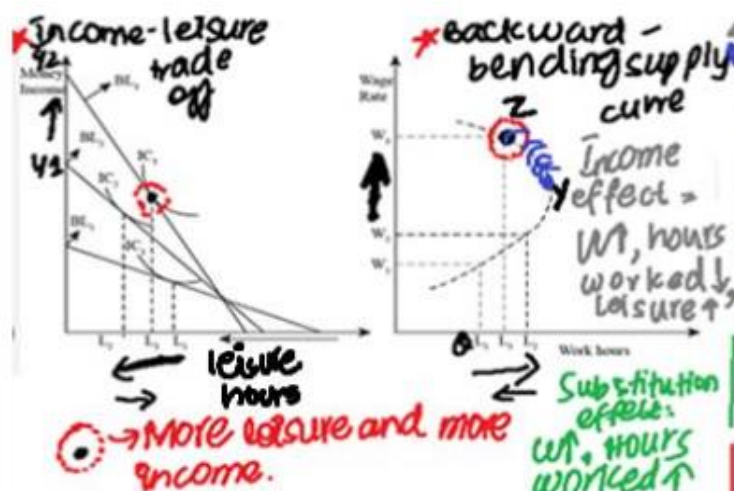
A. Marginal factor cost is the additional cost of employing one more unit of a factor.

B. Marginal revenue is the additional revenue from selling one more unit of output.

C. Marginal revenue product describes the additional revenue generated by adding one more unit of a factor of production (such as labor), while keeping all other factors constant.

D. The law of diminishing returns states that marginal product of labor will eventually decline over time as the quantity of labor employed increases given a fixed quantity of capital in the short-run, but the definition doesn't describe the law itself.

17-A:



→ More leisure and more income.

Substitution effect: $W \uparrow$, hours worked $\uparrow$

Wage $\uparrow$, Supply $\downarrow$ (hours worked)

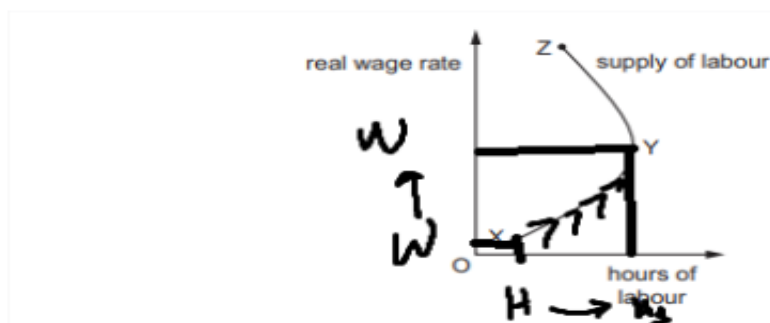
But as the wages increase further from W_2 to W_3 , and the target income is achieved, this means that the income effect will be negative. This will subsequently lead to an individual to contract some of their working hours from QL_2 to QL_3 , and enjoy a greater level of income from Y_1 to Y_2 , along with a greater leisure hours, shown through an increase in leisure hours enjoyed from L_2 to L_3 , provided that the decrease in working hours is proportionately lesser than the increased wage rate. Similarly here also the substitution effect will be positive, because like always an increase in wages will increase the incentives to work because the opportunity cost of leisure increases, but the "negative income effect" will outweigh the "positive substitution effect", and so the overall wage effect will be "negative". This results in a backward bend in the "labour supply curve".

-ve Income effect > +ve Substitution effect → Wage effect - negative

Wage effect → Wage $\uparrow$, Hrs $\downarrow$ Leisure $\uparrow$ extra leisure is preferred to work (4-2)

A – The reason why it is correct is explained in the diagram above.

B - Extra work is preferred to leisure between X to Y . This can be seen in the diagram below:



C – The working hours contracting between Y and Z is not a result of the job requiring more specific skills but is a result of an individual preferring extra leisure to work.

D – Although the pes of the final product will affect the pes of labor, but this has nothing to do with a backward bend in the supply curve.

18-B:

Economic migrants moving from rural to urban areas in developing countries can create benefits in both areas:

Rural areas:

- Receive financial help from urban workers, which can boost local economies and improve living standards.

Urban areas:

- Provide necessary labor for industries and services, contributing to urban economic growth and development. This influx of labor in urban areas may bring about a downward pressure on wages too.

Here's why the other options are incorrect:

A. Improving land ownership systems is not a direct benefit of migration, and urban workers may not necessarily become better educated.

C. While migration can reduce population pressure on resources in rural areas, living in informal settlements in urban areas is not a benefit.

D. Migration may not necessarily solve unemployment problems in rural areas, and obtaining loans from moneylenders can lead to debt traps.

19-B:

A – A decrease in the savings ratio or the proportion of incomes that is saved will mean that there would be less funds available for investment purposes in future which will hinder economic growth and development in the long-run.

B - An increase in investment in human capital is likely to increase potential economic growth in the long-run because an improvement in the quality of labor will boost the Long-run Aggregate Supply of the economy.

By enhancing productivity and efficiency, increasing innovation and entrepreneurship, it results in a reduction in poverty, as a result of which, the well-being of the entire society increases or in other words there is economic development. This is why, this becomes correct.

C - The depletion of non-renewable resources would lead to resource scarcity and hinder economic growth. As resources become scarce, costs increase, because of which, Aggregate supply decreases and subsequently cost push inflation occurs.

D - The greater use of compulsory overtime working would lead to burnout and decreased motivation, negatively impacting physical and mental health. While it may provide short-term productivity gains, it would decrease productivity and efficiency in the long run, hindering economic growth and development.

20- C:

20 The table gives population data from the United Nations.

	low-income countries	lower-middle-income countries	upper-middle-income countries	high-income countries
rate of population change (% per year)	2.7	1.5	0.8	0.5
life expectancy (years)	61	67	74	80
death rate (per 1000 population)	9	8	7	8
under-5 mortality (per 1000 live births)	90	59	18	6

Handwritten notes:
 - An arrow points from 2.7 to 0.5 with the note "population growth rate ↓".
 - An arrow points from 61 to 80 with the note "life expectancy ↑".
 - A circle around 7 and an arrow pointing to 8 with the note "death rate ↑".
 - An arrow points from 90 to 6 with the note "Infant mortality ↓".

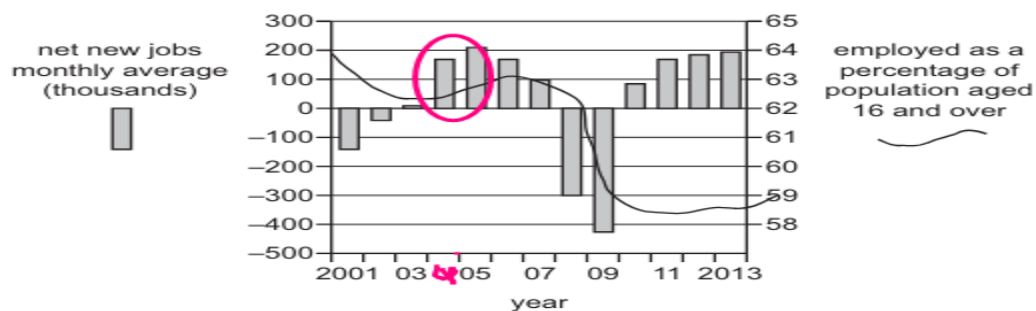
Expected trend:

As a country develops or in other words moves from low income to high income, the population growth rate is expected to fall. Additionally life expectancy is expected to rise, alongside which, infant mortality rate is expected to fall. All these indicators mentioned in the table "represent" the expected trend.

On the contrary, it is expected that as a country develops, the death rate falls. But in the table, it is shown to increase from 7 to 8 as the country moves from upper middle income to high middle income, which represents that key C will be correct since it "does not represent the expected trend"

21-A:

A – It can be seen that in the years when the number of net new jobs was rising (between 2004-2005), the percentage employed rose too, thus making this correct.



B – Employed as a percentage of population aged 16 and over did not decline continuously; it rose in certain years as well. (mainly between 2004-2007)

C – Based on this data, we cannot deduce the years in which natural rate of unemployment was achieved.

D – Based on data in percentages, we cannot say if number of employed workers rose in all except four years.

22-A:

A - This indicates that the country is becoming less dependent on foreign aid as it's earning foreign exchange through exports, investing in machinery, and developing its industries.

This explains why key A will be correct.

B: Changing economic structure to qualify for a loan is not a sign of reduced dependence on aid.

C: A deficit in the balance of payments due to interest payments indicates continued dependence on foreign capital.

D: Local firms going out of business due to competition from a multinational indicates vulnerability that may increase dependence on aid.

23-D:

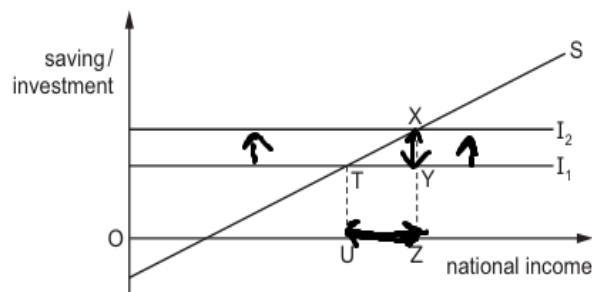
* gdp at factor cost
Indirect
add: taxes
less: subsidies
= gdp at
market
prices

gdp at market
prices
Add: subsidies
less: indirect taxes
= gdp at factor
cost

This explains why key D will be correct.

24-C:

The diagram shows the equilibrium levels of national income in a closed economy with no government, at two levels of investment, I_1 and I_2 .



Which ratio gives the value of the multiplier?

- A $\frac{TX}{UZ}$ B $\frac{UZ}{TX}$ C $\frac{UZ}{XY}$ D $\frac{XY}{UZ}$

2 sector multiplier

$$\frac{\Delta RNI}{\Delta AE} \quad \text{or} \quad \frac{1}{mps}$$

$$\frac{UZ}{XY}$$

25- A:

The correct answer is A: a reduction in taxation.

An upward shift of the consumption curve (also known as the consumption function) means that consumers are spending more at each level of national income. This can be caused by a reduction in taxation, as consumers have more disposable income to spend.

B: An increase in investment would shift the Aggregate expenditure curve, not the consumption curve.

C: An increase in national income would bring a movement along the consumption curve, not shift it upward.

D: An increase in savings would mean that consumers are spending less at each income level, which explains why this becomes incorrect.

26-D:

An increase in income level is most likely to result in an increase in the transactionary demand for money, as individuals have more funds to purchase goods and services.

- Price level rise (inflation): High inflation can also reduce the real value of money , which means that more money will be required to purchase the same level of goods and services. This can lead to an increased demand for money.
- Interest rate fall: A fall in interest rates would reduce savings as returns on saving fall, which means that consumption is likely to rise and subsequently demand for money will extend.

27-B: Keynesian theory advocates for increasing aggregate demand to prevent economic depression. This can be achieved through expansionary fiscal policy, where governments:

- Increase expenditure (government spending) to directly boost demand
- Decrease taxation to increase disposable income and encourage consumption.

Since option B represents reflationary fiscal policy or Keynesian demand management theory, it becomes correct.

A: Raising interest rates reduce Aggregate demand through contractionary monetary policy and it therefore contradicts Keynes' theory.

C: Encouraging consumer savings will reduce consumption and decrease aggregate demand, opposing Keynes' theory.

D: Restricting capital movement doesn't directly address aggregate demand, making it unrelated to Keynes' theory.

28-A:

A - A liquidity trap occurs when interest rates are so low that people and businesses hoard cash, rather than investing or consuming, even if the central bank increases the money supply. In this situation, monetary policy becomes ineffective, as the increased money supply is not being used to stimulate investment and growth. This occurs when money demand is perfectly elastic , because of which, interest rates remain unchanged.

B: Poverty trap refers to a situation where individuals are trapped in poverty due to various factors, including low income and lack of access to resources.

C: Marshall-Lerner condition is related to the J-curve effect in international trade. If the sum of ped of exports and imports is greater than 1, devaluation will bring about a balance of trade surplus.

D: The value of the multiplier being low would reduce the impact of the policy, especially if reduced interest rates boost autonomous investments after having reduced costs of borrowing, but it will not make it entirely ineffective.

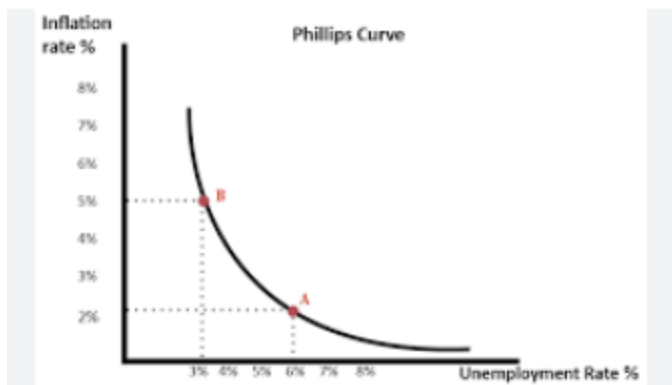
29-A:

The Phillips Curve shows a historical inverse relationship between inflation and unemployment, meaning:

- As unemployment falls, inflation tends to rise
- As unemployment rises, inflation tends to fall

This relationship suggests that there is a trade-off between inflation and unemployment, and that policymakers can use monetary and fiscal policies to manage this trade-off.

This can be seen in the diagram below:



Therefore, key A, becomes correct.

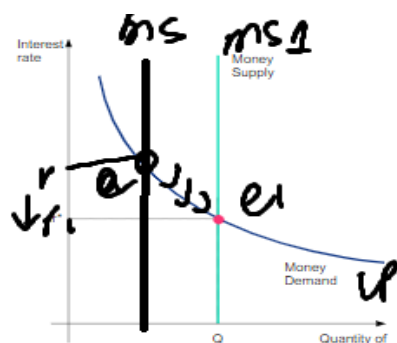
30-B:

B) If the country has a high marginal propensity to import (Mpm), a significant portion of the increased demand will be directed towards foreign goods, rather than domestic products. This can lead to a reduction in the value of the multiplier, which means that there will be smaller times increase in real national income arising from an increase in Aggregate expenditure.

A) A floating exchange rate gives the central bank autonomy to keep the interest rates low after they decrease due to a rise in money supply. This would have not been possible in a fixed exchange rate as the central bank may have to maintain a certain level of interest to be able to maintain the level of exchange rate.

C) A low marginal propensity to save represents that there must have been a higher marginal propensity to consume, which will boost the value of the multiplier. This means that there will be higher times increase in real national income arising from an increase in Aggregate expenditure.

D) The diagram and the explanation below explains how an interest inelastic demand for money will make monetary policy more effective in stimulating Aggregate demand, as it will lead to a fall in interest rates by a higher proportion upon a rise in money supply.



Therefore, monetary policy will be effective

* Initial money supply

↳ MS

* At equilibrium 'e', the money demand is relatively inelastic

* Now after an increase in money supply to MS1, the new equilibrium will be e1

* This represents that interest rate will fall by a higher percentage from r to r_1 .

